Date: November 7, 2026

To:
Editor-in-Chief
ACS Digital Medical Engineering
American Chemical Society
Email: `eic@dme.acs.org`

Subject: Submission of Original Research Article: *"Topologically Exact 3D/2D Line and Ray Clipping for Distributed CT and MRI Volume Computing: The Analytical Phase-Compensated Bresenham Algorithm"*

Dear Editor-in-Chief,

I am pleased to submit our original research manuscript entitled **"Topologically Exact 3D/2D Line and Ray Clipping for Distributed CT and MRI Volume Computing: The Analytical Phase-Compensated Bresenham Algorithm"** for consideration for publication as a regular **Article** in *ACS Digital Medical Engineering*.

Scope and Significance for *ACS Digital Medical Engineering*

High-resolution 3D volumetric medical imaging—including Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and PET/CT multi-modality scanning—demands spatial grid partitioning across distributed GPU clusters, MPI ranks, or tiled cache blocks to process massive voxel arrays (512^3 to 1024^3 voxels). Existing ray traversal pipelines clip continuous trajectories against sub-volume bounding boxes and round boundary intersections to integer coordinates to reseed standard Bresenham steppers.

In this work, we reveal that this conventional decoupled clipping strategy fundamentally resets fractional error accumulators, inducing a *Bresenham Phase-Shift Phenomenon* that corrupts voxel traversal paths across over 99.7% of boundary-crossing rays. In iterative tomographic reconstruction (SART, SIRT, MLEM), this introduces line-integral attenuation errors reaching up to $15.9\,\mathrm{cm}^{-1}$ and produces severe boundary streak and ring artifacts that impair diagnostic precision and mathematical convergence.

Key Advances Relative to the State-of-the-Art

To address this 60-year-old challenge at the interface of discrete geometry and computational biomedical imaging, this manuscript introduces the **Analytical Phase-Compensated Bresenham (AP-Bresenham)** algorithm:

1. **Closed-Form Diophantine Dynamics:** We derive the exact analytical equations governing multi-dimensional digital straight lines, establishing the Diophantine conditions of discrete ray stepping.

2. **Pure-Integer $O(1)$ Boundary Inversion:** We formulate exact $O(1)$ boundary inversion formulas directly in $\mathbb{Z}$, completely avoiding floating-point rounding errors and iterative searches.

3. **Analytical Historical Phase Injection:** We analytically restore the exact fractional accumulator state at any entry boundary in $O(1)$ time, guaranteeing 100% bit-level equivalence to monolithic traversal without requiring inter-node communication or ghost-cell buffers.

4. **Extensive Clinical Patient Validation:** Evaluated across synthetic Shepp-Logan phantoms, multi-compartment brain MRI models, and open clinical acquisitions (human abdominal CT and MNI-152 brain MRI) partitioned into up to 512 sub-volumes, AP-Bresenham completely eliminates seam artifacts, achieves exact Radon line-integral conservation (RMSE $= 0.0000, \text{PSNR} = \infty$), and delivers a 1.2×–1.6× setup speedup over floating-point clippers.

Prior Publication, Preprints, and Data Disclosure

- I confirm that this manuscript represents original work that has not been previously published in any peer-reviewed venue and is not under consideration for publication elsewhere.

- All software implementations (C++17 and Python), test suites (10 invariant suites, 11,000+ verification cases), reproducible benchmarks, and clinical validation scripts are open-source and publicly accessible on GitHub at: `https://github.com/hemanthmpkumar/AP-Bresenham`.

- Supporting Information containing extended mathematical proofs and algorithmic listings is provided as a separate accompanying document.

Suggested Independent International Reviewers

In accordance with the *ACS Digital Medical Engineering* submission guidelines, I provide five suggested independent expert reviewers from multiple countries who possess expertise in biomedical imaging, tomographic reconstruction, and computational geometry:

1. **Prof. David Coeurjolly**

   - *Affiliation:* CNRS, LIRIS, Université de Lyon, France
   - *Expertise:* Discrete Geometry, Digital Straight Lines, Digital Topology, Computational Imaging
   - *Email:* `david.coeurjolly@liris.cnrs.fr`

2. **Prof. Emil Y. Sidky**

   - *Affiliation:* Department of Radiology, The University of Chicago, USA
   - *Expertise:* Computed Tomography Reconstruction Algorithms, Forward Projectors, Artifact Modeling

- *Email:* sidky@uchicago.edu

3. **Prof. Gabor T. Herman**

   - *Affiliation:* Department of Computer Science, City University of New York (CUNY), USA
   - *Expertise:* Discrete Tomography, Algebraic Reconstruction Techniques (ART), Volume Computing
   - *Email:* gherman@gc.cuny.edu

4. **Prof. Vaclav Skala**

   - *Affiliation:* Department of Computer Science and Engineering, University of West Bohemia, Czech Republic
   - *Expertise:* Geometric Clipping Algorithms, Projective Coordinates, High-Performance Rasterization
   - *Email:* skala@kiv.zcu.cz

5. **Prof. Bruno Sixou**

   - *Affiliation:* INSA Lyon, CREATIS, CNRS UMR 5220, France
   - *Expertise:* Tomographic Image Reconstruction, Discrete Optimization, Radiological Path Traversal
   - *Email:* bruno.sixou@creatis.insa-lyon.fr

None of the proposed reviewers have been collaborators, co-authors, mentors, or mentees of the author within the past five years.

Thank you very much for your time and consideration of our manuscript.

Sincerely,

Hemanth Manchabale Papachappa
Independent Researcher
Seattle, Washington, USA
Email: hemathmpkumar123@gmail.com
ORCID: 0009-0007-1398-9451